

SUPERPOSITION OF CYCLOTRON LINES IN ACCRETING X-RAY PULSARS. I. LONG SPIN PERIOD

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ABSTRACT

I model cyclotron lines by calculating a superposition of a large number of cyclotron line spectra emerging from different heights of a line-forming region of a non-uniform density and temperature of an electron–proton plasma threaded by a dipole magnetic field. In this paper, I mainly discuss the properties of the cyclotron lines in accreting X-ray pulsars with long spin periods (~ 100 – 800 s). My simplified model succeeds in reproducing asymmetric line profiles, anharmonic ratios of line energy, relationship between the width and the depth of the line, and correlation between cyclotron line energy and X-ray intensity in the main pulse. In addition, I demonstrate that a shallower and broader fundamental (= first harmonic) line and a deeper second harmonic line as observed in accreting X-ray pulsars such as Vela X–1 and A0535+26 can be produced via a superposition of a large number of cyclotron lines formed at numerous different heights in a line-forming region with a dipole magnetic field. In this case, the ratio of the peak energy of the second harmonic absorption line to the fundamental can be larger than harmonic ratio 2. Furthermore, the width of the fundamental line is comparable to or larger than that of the second harmonic. These characteristics of the cyclotron lines are consistent with those observed in Vela X–1 and A0535+26.

Key words: stars: atmospheres – stars: magnetic field – stars: neutron – X-rays: binaries

1. INTRODUCTION

Cyclotron lines in the spectra of X-ray accreting pulsars provide information on the geometry, the electron temperature, and density, as well as the strength of the magnetic field (B -field) of a line-forming region above the surface of a neutron star (NS). The cyclotron lines have been discovered in the spectra of more than 15 accreting X-ray pulsars (e.g., Makishima et al. 1999; Coburn et al. 2002). Most efforts to calculate theoretical cyclotron lines have been performed in a line-forming region with a constant temperature and density of an electron–proton plasma permeated by a uniform magnetic field (Nagel 1981a, 1981b; Mészáros & Nagel 1985; Wang et al. 1988, 1989; Isenberg et al. 1998; Araya & Harding 1999; Araya-Góchez & Harding 2000). However, Bulik et al. (1992, 1995) pointed out that even maximal pole-on Doppler broadening is unable to fit the line profiles with a single field strength or a dipole distribution at a constant radius in 4U1538–52 and Vela X–1. Nishimura (2003) demonstrated that the variations on the strength of a magnetic field affect the profile of the cyclotron line. Nishimura (2005) also calculated cyclotron lines assuming a strong variation on field strength with a distance from an emission region. This model is able to account for deviations from the classical harmonic ratio of the centroid energies of the cyclotron lines of the second harmonic to the fundamental (= first harmonic). Thus, a non-uniformity in the line-forming region can affect certain characteristics of the cyclotron line. Nishimura (2008) models cyclotron lines by calculating a superposition of the cyclotron line spectra emerging from a large number of individual line-forming regions in a cylindrical geometry. A uniform density and temperature of an electron–proton plasma threaded by a constant magnetic field is taken to be in the individual line-forming regions, while the values of the density, the temperature, and the magnetic field are assumed to change with the heights of the individual line-forming regions from the surface of NS. This model succeeded in reproducing broader and shallower cyclotron lines as commonly observed in accretion-powered X-ray pulsars. However, this model cannot sufficiently reproduce the behavior of the cyclotron lines as can be seen in ac-

creting X-ray pulsars with long spin periods (~ 100 – 800 s) as follows: first, the ratio of the line energy of the second harmonic to the fundamental can be more than 2 such as Vela X–1 (Kreykenbohm et al. 2002). Second, the second harmonic absorption line can be deeper, while the fundamental line can be quite shallow in several X-ray pulsars, such as Vela X–1 and A0535+26, where it had been difficult to detect the fundamental line whereas the second harmonic line was detectable (e.g., Orlandini et al. 1998). In the case of Vela X–1, very shallow fundamental lines have been detected in recent observations (e.g., Kreykenbohm et al. 2002). In the case of A0535+26, the fundamental lines shallower than the second harmonic lines were detected in the 2005 August/September normal (type I) outburst (e.g., Caballero et al. 2007). Third, the line width of the fundamental line can be comparable to or more than that of the second harmonic line, such as Vela X–1 and A0535+26. This is contradicted with the relationship predicted by thermal Doppler width.

In accreting X-ray pulsars with long spin periods (~ 100 – 800 s), the fundamental line tends to be even shallower, while it tends to be deeper in those with short spin periods (~ 1 – 10 s; Coburn et al. 2002). In theoretical model for non-rotating or very slowly rotating (period > 100 s) X-ray sources, Arons & Lea (1980) found that

$$A_{\text{cap}} \cong 4 \times 10^{11} (R_*/10 \text{ km})^3 \mu_{30}^{-3/2} L_{37}^{-3/5} \text{ cm}^2, \quad (1)$$

under the assumption that spherically accreting plasma enters the magnetosphere via the Rayleigh–Taylor instability. Their model is able to yield a larger polar cap area, especially for lower luminosity. This model may be applied to accreting X-ray pulsars with long spin periods (~ 100 – 800 s) and possibly those with relatively long spin periods (~ 10 – 100 s) discussed in this paper.

Thus, these recent observational results motivated us to perform the calculations of a superposition of the cyclotron line spectra emerging from different heights or sites with different magnetic field strengths in a large line-forming region. The paper is organized as follows. In Section 2, I discuss the

structures of line-forming region I employ in my calculations. Radiative transfer calculations are briefly explained in Section 3. In Section 4, I present the spectra calculated by my model. I discuss the influence of a superposition of a large number of lines on the properties of resulting cyclotron lines and compared my numerical results to observations for certain accretion-powered pulsars in Section 5. In Section 6, I draw my conclusions and summarize my results.

2. STRUCTURES OF THE LINE-FORMING REGIONS

In the present paper, I examine the effect of a superposition of a large number of cyclotron lines emerging from different heights of a line-forming region with variations in the strength of a magnetic field, the temperature, and the density of an electron-proton plasma on the property of the resulting lines. In this work, I assume that a continuum-forming region, in which a continuum spectrum mainly formed, is located below the line-forming region as shown in Figure 1(a). I use a power-law ($\propto E^{-1}$) energy spectrum as a continuum spectrum. Thus, the line-forming region is assumed to be illuminated by a power-law energy continuum spectrum from below, unlike Nishimura (2008), who considered a cylindrical geometry with an emission region along the cylindrical axis. Thus, I consider a cylindrical, plane-parallel geometry with a height z up to a few kilometers from the surface of the NS for the line-forming region, as shown in Figure 1(a). This line-forming region is thick to cyclotron resonant scattering and thin to continuum scattering. This model may correspond to a shape like an accretion mound with emission regions of different heights, as shown in Lyubarskii & Syunyaev (1988), rather than a cylindrical geometry.

I use 1–0 geometry in order to calculate the cyclotron lines formed in different heights of the line-forming region as shown in Figure 1(b). Here, the 1–0 geometry is a slab illuminated from below as defined in Isenberg et al. (1998), where the photons emerging from the bottom of the slab are assumed to return to the surface of NS where they are thermalized. Thus, the bottom of each slab is located at the same height. As a result, photons injected from the bottom of the line-forming region near the wall emerge from a lower altitude of an accretion column, while other photons injected from the bottom of the line-forming region far from the wall emerge from a higher altitude of an accretion column. Moreover, the gradients in the magnetic field, the temperature, and the density of the plasma in each slab with the height are taken into account, unlike the work of Nishimura (2008), in which their values in each region were taken to be constant. Once the distributions of the temperature, the density, and the magnetic field in each slab are determined, I can calculate the cyclotron line emerging from each slab with magnetic Feautrier radiative transfer code.

I assume the slab geometry threaded by a dipole magnetic field oriented along the slab normal, since Nakajima et al. (2006) succeeded in explaining the change of fundamental resonance energy with luminosity in X0115+63 under the assumption of a dipole magnetic field configuration. Here, the magnetic field is assumed to be a dipole, $B = B_s (\frac{R_{NS}}{R_{NS}+z})^3$, where R_{NS} is the radius of NS and B_s is the magnetic field at the surface of NS. The temperature is also approximated by $kT \sim \hbar\omega_{cyc}/4$, which is derived by Lamb et al. (1990) from a balance of the cooling and heating that arises from cyclotron resonant and nonresonant scattering. The density distribution can influence the profile of a cyclotron line, especially in a line-forming region with a non-uniform magnetic field. Because of the lack of a density

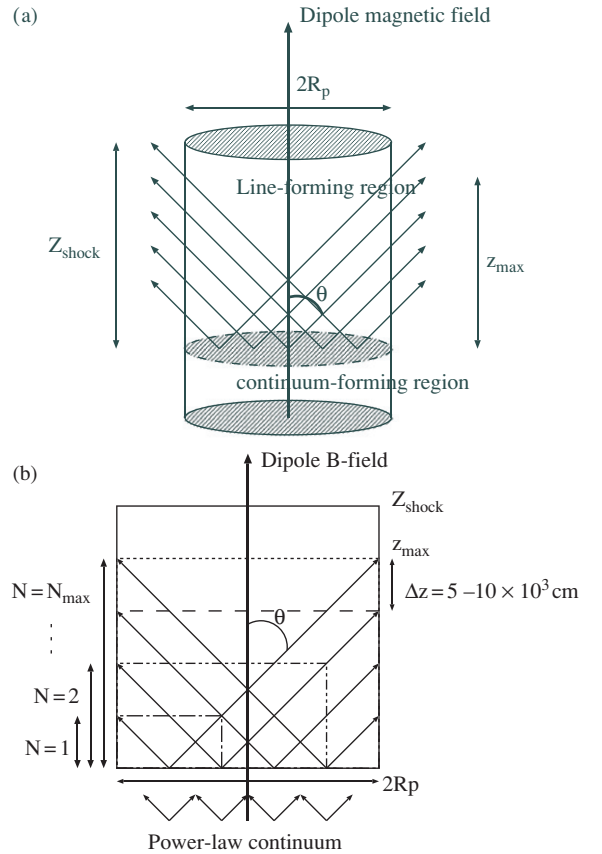


Figure 1. (a) Line-forming region illuminated from below is a cylindrical plane-parallel geometry. Continuum-forming region is assumed to be located below the line-forming region. (b) Cyclotron lines emerging from different heights of the line-forming region are calculated in the plane-parallel geometry of different heights. Regions of different heights denoted by dotted, dashed, two dots-dashed, and dot-dashed lines show slabs used in calculations.

distribution model in a line-forming region, in my calculations I make use of an analytical density profile, which is derived from an analytical velocity profile via the equation of continuity in a magnetically confined accretion column, at a given height z of the line-forming region (Becker 1998):

$$\rho(z) \propto \left[1 - \left(\frac{7}{3} \right)^{-z/z_{shock}} \right]^{-1}, \quad (2)$$

where the shock height z_{shock} is a free parameter. In these assumptions, my model would be appropriate for accreting X-ray pulsars with luminosity $\lesssim 10^{36} - 10^{37} \text{ erg s}^{-1}$, while the model I adopt for density distribution would be appropriate for higher luminosity $\gtrsim 10^{36} - 10^{37} \text{ erg s}^{-1}$. However, the model of the density distribution succeeded in reproducing a continuum spectrum in 4U 0352+309 with quite a low luminosity $\sim 10^{34} \text{ erg s}^{-1}$ (Becker & Wolff 2005). The model of the density distribution may therefore be able to be applied to accreting X-ray pulsars with quite a low luminosity, $\sim 10^{34} \text{ erg s}^{-1}$.

The polar cap radius is taken to be $R_p \sim (1-2) \times 10^5 \text{ cm}$, because such a larger polar cap can be formed in accreting X-ray pulsars with a long spin period for lower luminosity, $L_x \lesssim 10^{37} \text{ erg s}^{-1}$, from Equation (1). The top of each slab z_{top} is assumed to be located at different heights by $(5-10) \times 10^3 \text{ cm}$, in which the difference of the height corresponds to a resonance energy of $\sim 1 \text{ keV}$, as shown in Figure 1(b). The bottom of the line-forming region is assumed to be constant at a height

of 1×10^4 cm from the surface of the NS. In X0115+63, the height of the line formation is reported to converge to a finite value of $0.03 R_{\text{NS}} - 0.07 R_{\text{NS}}$ for the luminosity $L_x \lesssim 10^{37} \text{ erg s}^{-1}$ (Nakajima et al. 2006). It is possible, however, that the height of the bottom is lower than that for an accreting X-ray pulsar with a short spin period. This is because the density of an accretion column in an accreting X-ray pulsar with a long spin period is expected to be lower for a larger polar cap radius due to $\rho \propto L_x / \pi R_p^2$. As a result, the height of the top of the continuum-forming region can be lower. In addition, the line energy of the fundamental tends to be formed around upper region of the line-forming region as shown in Nishimura (2003). Thus, in this work, the bottom of the line-forming region is taken to be lower than $z = (3-7) \times 10^4$ cm for $R_{\text{NS}} = 10^6$ cm. Thus, I calculate the cyclotron lines emerging from a large number of slabs and then perform a superposition of a large number of cyclotron lines.

The dimensions of the polar cap region can affect the number of superposing lines. I will discuss how it influences the resulting cyclotron line. I use a power-law ($\propto E^{-1}$) energy continuum for simplicity, although the effect of photon spawning from Raman scattering at higher harmonics (production of lower harmonic photons from a higher harmonic photon) on the depth and shape of fundamental line can be influenced by the continuum shape (Schönherr et al. 2007). However, in the present work, I focus on the influence of a superposition of a large number of lines on the properties of the resulting cyclotron lines, because it is expected to dominate over photon spawning when the superposition of a large number of lines occurs. I will examine the influence of the shape of certain observed spectra on the cyclotron line in the forthcoming paper.

Thus, this study will be able to offer guides to understanding the property of the lines formed in an actual line-forming region, such as an accretion column, an accretion halo (thin gas region outside an accretion column), the free-fall region of an accretion column, or a radiation-supported scattering atmosphere.

3. RADIATIVE TRANSFER

I use the magnetic Feautrier radiative transfer code developed in Nishimura (2008). The basic idea is the same as was described in Nishimura (2008), except that the emission region of the line-forming region is different, and the temperature, the density, and the B -field change with altitude in each slab as described in Section 2. The cyclotron line is modeled by superposing N original cyclotron lines calculated with 120 frequencies. In this work, $N_{\text{max}} = 7-38$. The original cyclotron line emerging from each slab is calculated for scattering through several choices of the viewing angle in a two-stream formalism. The total photon number density, U_{total} , is computed as the sum of photon number densities, U_{D_i} , emerging from the N slabs,

$$U_{\text{total}} = \frac{1}{N} \sum_{i=1}^N U_{D_i}. \quad (3)$$

An initial power-law photon number density, $U(E) \propto \omega^{-1}$, is injected into each slab. I include excitation up to three harmonics in Landau levels and photon spawning followed by a decay of these levels in cyclotron resonant scattering. I modified the Feautrier radiative transfer code given in Nishimura (2008) as follows: I take the distribution of electron momenta along the field, $f(p)$, to be a one-dimensional relativistic Maxwellian

(Wang et al. 1993),

$$f(p)dp = N_e \exp\left(-\frac{\sqrt{1+p^2}-1}{T}\right) dp. \quad (4)$$

The cross-section of cyclotron resonant scattering for relativistic electrons and $B \sim B_{\text{crit}}$ by summing over the final spin state and averaging over the photon polarization is given by (Harding & Daugherty 1991)

$$\sigma_{\text{abs}}^n(\theta) = \frac{\alpha \pi^2 \hbar^2 c^2}{E_n} \frac{\Gamma_n/2\pi}{(\omega - \omega_n)^2 + \Gamma_n^2/4} \times \frac{e^{-Z} Z^{n-1}}{(n-1)!} \left[(1 + \cos^2\theta) + \frac{Z}{n} \sin^2\theta \right], \quad (5)$$

where

$$Z \equiv \frac{\omega^2 \sin^2\theta}{2m^2 B'}, \quad (6)$$

$$E_n = (m^2 + \omega^2 \cos^2\theta + 2nB'm^2)^{1/2}, \quad (7)$$

and

$$\hbar\omega_n = mc^2[(1 + 2nB'\sin^2\theta)^{1/2} - 1]/\sin^2\theta, \quad (8)$$

where the width Γ_n is the decay rate from the Landau excited level n , θ is the angle between the photon momentum and the magnetic field, $B' = \frac{B}{B_{\text{crit}}}$. I use the accurate approximate form as the cross-section for cyclotron resonant scattering,

$$\sigma_{\text{sc}}^n \simeq \frac{\sigma_{\text{abs}}^n}{L_n} \frac{E_n^2 \Gamma_n / 2\pi m^2}{(\omega + \omega^2 \sin^2\theta / 2m - nB'm)^2 + (E_n \Gamma_n / 2m)^2}, \quad (9)$$

where L_n is the Lorentz profile (Harding & Daugherty 1991).

4. RESULTS

In Section 4.1, superposed cyclotron lines are computed using one angle. I demonstrate how they change with the number of superposed lines. The characteristics of the line depth, width, and energy as well as the harmonic ratios of the line energies are shown. In Section 4.2, angle-dependent spectra of superposed cyclotron lines are computed using five angles. I demonstrate how the characteristics of the superposed cyclotron line change with viewing angle or polar cap radius.

4.1. Characteristics of Superposed Cyclotron Lines

I first calculate superposing cyclotron lines using an angle of $\mu \equiv \cos\theta = 0.57$, in which the resulting spectrum approximately represents angle-averaged spectrum. The cyclotron energy at the bottom of the line-forming region is taken to be $\hbar\omega = 30$ keV. Here, the line-forming region covers the cyclotron energy from ~ 17 to ~ 30 keV via a dipole law, which would correspond to the case of VelaX-1. Figure 2 shows superposed cyclotron lines for the numbers of superposed lines, $n_s = 1, 4, 8, 12, 16, 19$, which can correspond to the polar cap radii $R_p \sim 9 \times 10^3, 3 \times 10^4, 6 \times 10^4, 9 \times 10^4, 1.15 \times 10^5$, and 1.4×10^5 cm, respectively. In my model, the size of the polar cap radius R_p is given by

$$R_p = (z_{\text{max}} \tan\theta)/2 = z_{\text{max}} \sqrt{1 - \mu^2} / 2\mu \sim 0.72 z_{\text{max}} \quad (10)$$

for a possible maximum height z_{max} of the line-forming region and a given viewing angle as shown in Figure 1(b). Thus, the

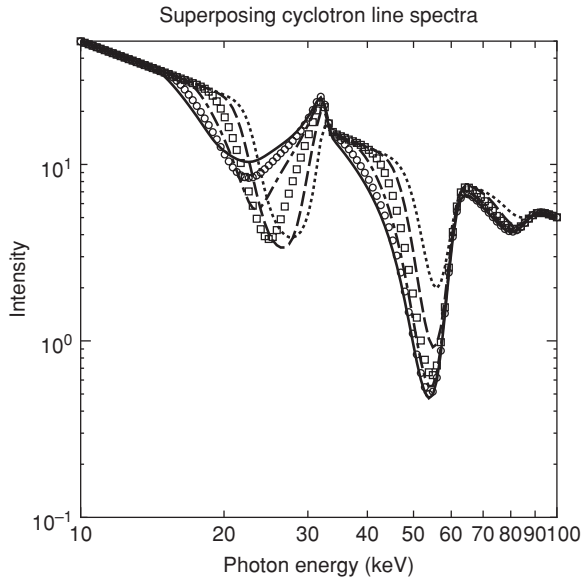


Figure 2. Superposition of cyclotron lines for the number of the superposed lines $n_s = 1$ (dotted line), 4 (dashed line), 8 (open squares), 12 (dot-dashed line), 16 (open circles), and 19 (solid line). The viewing angle is taken to be $\mu = 0.57$. The shock height is $z_{\text{shock}} = 2 \times 10^5$ cm. The electron number density ranges from $N_{e,21} \sim 1.7$ to $N_{e,21} \sim 11.4$, where $N_{e,21} \equiv N_e/(10^{21} \text{ electrons cm}^{-2})$.

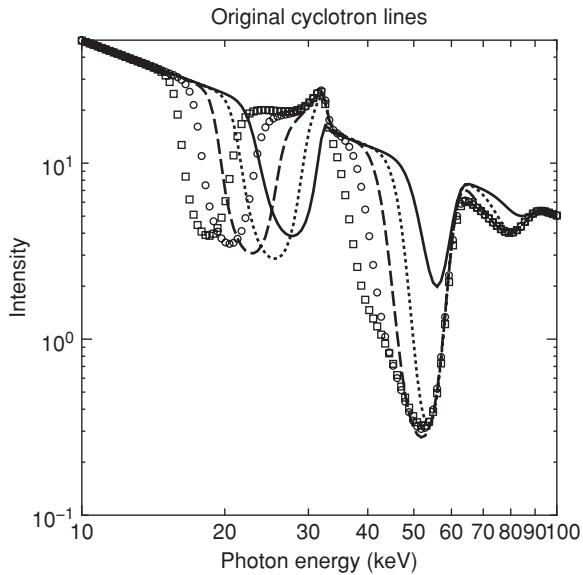


Figure 3. Original lines emerging from several different heights in the line-forming region, for the heights, $z = 1.0 \times 10^4$ (solid line), $z = 4.0 \times 10^4$ (dotted line), $z = 8.0 \times 10^4$ (dashed line), $z = 1.2 \times 10^5$ (open circles), and $z = 1.6 \times 10^5$ (open squares) cm.

broader the polar cap, the larger the number of the superposition of the lines tends to be.

The fundamental line becomes broader and shallower with the increase in the number of superposed lines, whereas the second harmonic line becomes deeper and deeper. Figure 3 shows that each original fundamental line emerging from a different height is different in peak energy, which corresponds to the strength of the magnetic field near the altitude photons escape from the line-forming region. The fundamental line is formed via multiple scattering so that its peak energy, $E_{\text{cyc},1}$, tends to be formed around the sites where photons escape from the line-forming region (e.g., the wall of an accretion column) despite lower density. This is because a large number of photons

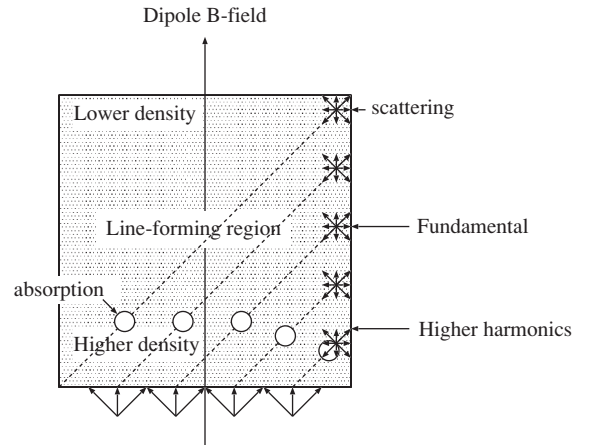


Figure 4. Schematic depiction of the sites forming the peak energies of cyclotron lines. Points denoted by a group of eight arrows, which illustrate scattering event, indicate the sites where the peak energies of a number of original fundamental lines are formed. Circles, which illustrate absorption event, indicate the sites where the peak energies of a number of original higher harmonic lines are formed. “Fundamental” and “Higher harmonics” denote roughly the sites around which the peak energies of the fundamental and higher harmonic lines, respectively, are formed as a result of the superposition of the lines.

scattered in the upper regions of the line-forming region tend to refill the absorption feature formed in the lower regions of the line-forming region, transferring to different energies (Nishimura 2003). Thus, the peak energy of the fundamental absorption line tends to indicate the value of the B -field near the altitude at which photons escape from the line-forming region, as illustrated in Figure 4. In contrast, each original second harmonic line indicates nearly the same peak energy, as shown in Figure 3. This is because the formation mechanism via almost pure absorption process by Raman scattering generates the second harmonic lines primarily around the bottom of the line-forming region due to the highest density, as illustrated in Figure 4. Consequently, the resulting second harmonic line tends to become deeper in spite of the increase in the number of the superposed lines, whereas the fundamental tends to become shallower and broader. In addition, the peak energies of the second and third harmonic lines nearly remain the same despite the increase in the number of the superposed lines, while the peak energy of the fundamental line shifts noticeably to lower energy. As a result, the ratio of the peak energy of the second harmonic line to the fundamental ($E_{\text{cyc},2}/E_{\text{cyc},1}$) tends to be larger than the harmonic ratio 2, whereas that of the third harmonic line to the second ($E_{\text{cyc},3}/E_{\text{cyc},2}$) remains 1.5.

Mészáros & Nagel (1985) roughly predicted the FWHM of the line

$$\Delta\omega_{\text{FWHM}} \sim \omega_c \left(8 \ln(2) \frac{kT}{mc^2} \right)^{1/2} |\cos \theta|. \quad (11)$$

In my calculations, however, the width of the fundamental line $W_{1\text{st}}$ can be comparable to or larger than that of the second harmonic $W_{2\text{nd}}$ as a result of the superposition of a large number of lines ($n_s \gtrsim 18$), in contradiction to the relationship $W_{2\text{nd}} \sim 2W_{1\text{st}}$ predicted by theoretically Doppler broadening (Equation (11)). This is because the second harmonic line width does not increase as much as that of the fundamental line due to the original second harmonic lines with nearly the same peak energy. In addition, the superposition of a large number of lines ($n_s \gtrsim 18$) produces an asymmetric line profile, which is shallower toward lower energies. This is because the maximum

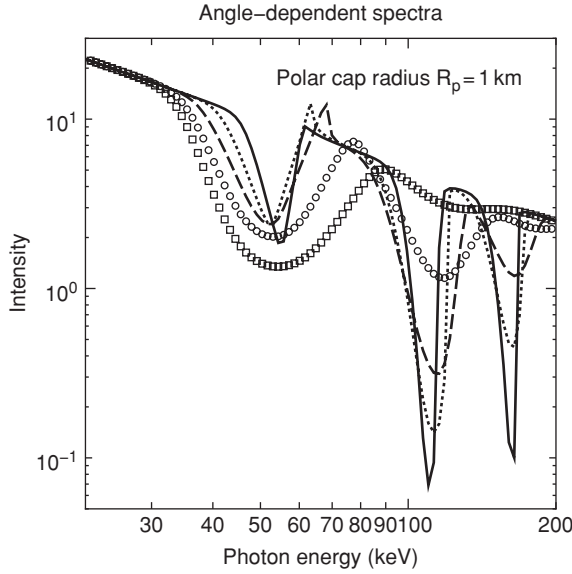


Figure 5. Angle-dependent spectra formed via the superposition of the lines at five viewing angles, $\mu = 0.18$ (solid line), 0.34 (dotted line), 0.52 (dashed line), 0.79 (open circles), and 0.96 (open squares). Both the polar cap radius R_p and the shock height z_{shock} are assumed to be 1×10^5 cm. The line-forming region covers the cyclotron energy from ~ 45 to ~ 60 keV via a dipole law. The electron number density ranges from $N_{e,21} = 1.4$ to $N_{e,21} = 12.2$.

Table 1
Harmonic Ratio of Line Energy and Width

$E_{\text{cycl},2}/E_{\text{cycl},1}$	$E_{\text{cycl},3}/E_{\text{cycl},2}$	$W_{2\text{nd}}/W_{1\text{st}}$
$\gtrsim 2$	~ 1.5	~ 1

overlapping area of a large number of original absorption lines is located around higher energies, as seen in Figure 3. The ratio of the line energies and widths of the second harmonic to the fundamental and the ratio of the line energies of the third harmonic to the second as a result of the superposition of a large number of lines are summarized in Table 1.

4.2. Angle-dependent Spectra

I consider angle-dependent spectra for five angles, $\mu = 0.18, 0.34, 0.52, 0.79$, and 0.96 ($\theta \sim 79^\circ, 70^\circ, 58^\circ, 37^\circ$, and 16°), as shown in Figure 5. Here, the line-forming region covers the cyclotron energy from ~ 45 to ~ 60 keV via a dipole law, which would correspond to the case of A0535+26. From Equation (10), in my model, the larger the viewing angle is with respect to the magnetic field, the smaller the number of the superposed lines tends to be in a small polar cap region of $R_p \sim 1 \times 10^5$ cm. The number of the superposed lines n_s are 7, 14, 18, 18, and 18 for angles $\mu = 0.18, 0.34, 0.52, 0.79$, and 0.96 , respectively. Thus, the peak energies of broader lines at the fundamental tend to be lower as a consequence of the superposition of a larger number of lines. However, the peak energy of the line at the smallest angle of $\mu = 0.96$ appears to be somewhat higher than that at an angle of $\mu = 0.79$. This is because the overlapping area of the original lines is larger for a broader line so that the peak energy of the line remains somewhat higher: the peak energy of a narrower line tends to indicate lower energy when the number of the superposing lines is same. On the other hand, the peak energy of the second harmonic line remains nearly the same in spite of the superposition of a large number of lines. Consequently, the ratio $E_{\text{cycl},2}/E_{\text{cycl},1}$ tends to be more than 2 for smaller viewing angles whereas

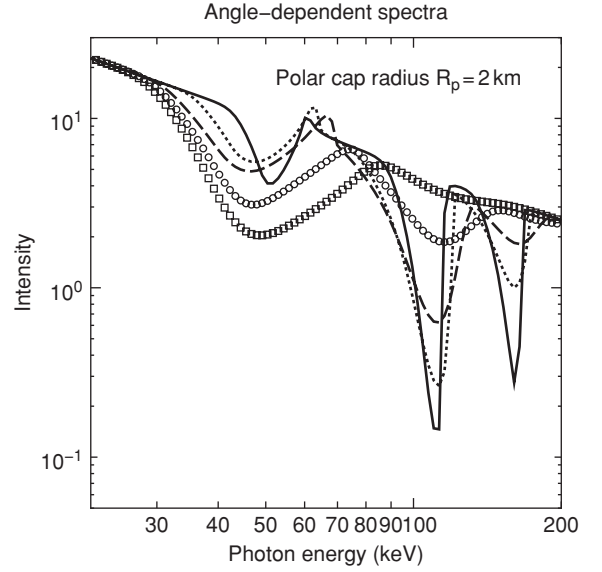


Figure 6. Same as Figure 5 but the polar cap radius R_p and the shock height z_{shock} are assumed to be 2×10^5 cm. The electron number density ranges from $N_{e,21} = 0.67$ to $N_{e,21} = 8.0$.

it remains nearly 2 for larger viewing angles as a result of the superposition of a smaller number of lines. The second harmonic line, however, tends to be even shallower for a smaller angle, since the scattering profiles of higher harmonics are dominated by a $\sin^{2(n-1)}\theta$ factor. Moreover, for smaller viewing angles, the line width of the fundamental $W_{1\text{st}}$ can be comparable to or larger than that of the second harmonic $W_{2\text{nd}}$, in contradiction to the relationship $W_{2\text{nd}} \sim 2W_{1\text{st}}$ predicted by theoretically Doppler broadening (Equation (11)). On the other hand, for larger viewing angles, the line width of the second harmonic is nearly twice times that of the fundamental, as expected by Equation (11).

The line depth tends to be deeper for a smaller viewing angle. This is because the thermal Doppler width becomes broader at smaller angle. The overlapping area of the original lines is therefore larger for smaller angle so that the resulting line remains deeper for smaller angle. At $\mu = 0.18$, however, the line depth remains somewhat deeper because of the superposition of a small number of lines.

Figure 6 is the same as Figure 5 but the polar cap radius $R_p \sim 2 \times 10^5$ cm, in which the density is lower because of $\rho \propto L_x/\pi R_p^2$. In such a large polar cap region, the superposition of a large number of fundamental lines in all viewing angles occurs. The number of the superposed lines n_s are 15, 29, 38, 38, and 38 for angles $\mu = 0.18, 0.34, 0.52, 0.79$, and 0.96 , respectively. As a result, the fundamental lines in all viewing angles become shallower and broader than $R_p \sim 1 \times 10^5$ cm. A broader fundamental line tends to be deeper for the angle from $\mu \sim 0.34$ to 1. On the other hand, the second harmonic lines remain relatively deeper, although they become somewhat shallower than the case for $R_p \sim 1 \times 10^5$ cm because of lower density $\rho \propto L_x/\pi R_p^2$. The peak energies of the fundamental lines shift to lower energies in all viewing angles, while those of higher harmonics nearly remain the same. As a result, the harmonic ratios $E_{\text{cycl},2}/E_{\text{cycl},1}$ become more than 2 at all viewing angles. Moreover, for most of viewing angles, $W_{1\text{st}} \gtrsim W_{2\text{nd}}$. The characteristics of the cyclotron lines for the smallest and largest viewing angles $\mu = 0.18$ and 0.96 in phase-resolved spectra for $R_p = 1 \times 10^5$ cm and 2×10^5 cm, are summarized in Table 2.

Table 2
Angle-dependent Spectra

R_p	μ	$E_{\text{cycl},2}/E_{\text{cycl},1}$	$W_{2\text{nd}}/W_{1\text{st}}$
1 km	0.96	>2 (~ 2.29)	~ 1
1 km	0.18	~ 2 (~ 2.02)	~ 2
2 km	0.96	>2 (~ 2.47)	~ 1
2 km	0.18	>2 (~ 2.24)	~ 1

5. DISCUSSION

In this section, I will discuss variations on the line depth and width with luminosity as well as the relationship between E_{cycl} and the intensity in the main pulse of certain accreting X-ray pulsars. Cyclotron lines will be discussed in accreting X-ray pulsars with long spin periods (~ 100 – 800 s) in Section 5.1. In Section 5.2, I will discuss cyclotron lines in accreting X-ray pulsars with relatively long spin periods (~ 10 – 100 s). In Section 5.3, comprehensive comparisons are performed.

5.1. Long Spin Periods (~ 100 – 800 s)

For an accreting X-ray millisecond pulsar, Poutanen & Gierliński (2003) found that blackbody radiation is strongly beamed along the normal to the slab (a pencil-like emission pattern) while the Comptonized emission has a broader angular distribution, peaking at about 50° – 60° from the slab normal (a fan-like emission pattern). In this section, I discuss how the line depth and width change with luminosity for each accreting X-ray pulsar with a long spin period ($\gtrsim 100$ s), taking into account the patterns of pencil-like and fan-like beams shown in their model in addition to a pencil or a fan beam. The pencil-like and fan-like beams used in the present paper are illustrated in Figure 7. Moreover, I roughly explain the relationship between E_{cycl} and the intensity using the beam patterns. Note that the beam pattern would depend on the polar cap radius and the mass of NS as well as luminosity from the fact that critical luminosity is determined by (Becker 1998)

$$L_{\text{crit}} = 3.14 \times 10^{37} \left(\frac{M_{\text{NS}}}{M_{\odot}} \right) \left(\frac{R_p}{R_{\text{NS}}} \right) \text{ erg s}^{-1}, \quad (12)$$

where M_{NS} is the mass of the NS. Equation (12) indicates that the beam pattern of an accreting X-ray pulsar with a smaller polar cap radius is characterized by a fan beam for lower luminosity. In this section, I roughly discuss how the line depth and width change via a variation of an angular distribution of the emission with luminosity. I discuss the cyclotron lines in phase-averaged or phase-resolved spectra in the main pulse of accreting X-ray pulsars with long spin periods (~ 100 – 800 s).

5.1.1. Higher Luminosity $\sim 10^{37}$ erg s $^{-1}$

I will first consider accreting X-ray pulsars with a higher luminosity $\sim 10^{37}$ erg s $^{-1}$ which would be close to L_{crit} . In such a high luminosity, a fan-like beam is expected, as shown in Figure 7, so that the angle of the peak emission with respect to B -field tends to be larger. In this case, the fundamental line tends to be shallower and narrower in Figure 6, while the second harmonic line tends to be deeper. An angular distribution of the emission with the peak at a larger angle would tend to yield relatively clear double peaks in the pulse profiles around the energy of the fundamental cyclotron line, although it depends on the angle between spin axis and the line of sight or the viewing angle θ as well as the luminosity.

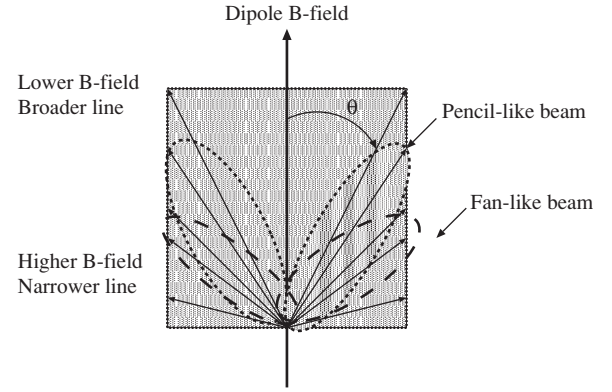


Figure 7. Schematic depiction of the relationship between the cyclotron line characteristics (energy and width) and the intensity of the emission in the pulse profile. Dashed and dotted lines indicate the angular distributions of the intensity. The solid lines represent angle-dependent propagation radiation through the line-forming region. A pencil-like beam (dotted lines) is relatively close to a pencil beam, while a fan-like beam (dashed lines) is relatively close to a fan beam.

A0535+26. In the case of A0535+26 with pulsation ~ 103 s, the luminosity reaches $\gtrsim 10^{38}$ erg s $^{-1}$ in a giant outburst in 1994 (Maisack et al. 1997). In this giant outburst, the fundamental line was too shallow to detect it as a line, while a second harmonic line could be detected. Grove et al. (1995) reported that the ratio of the depth of a 55 keV feature to a 110 keV is less than 1/3.5. On the other hand, in a normal (type I) outburst in 2005 August/September, a fundamental cyclotron line is relatively deeper and a second harmonic line is detected mainly in higher luminosity of the outburst (Caballero et al. 2007). This trend can be explained by the idea that the angle of the peak emission becomes larger as luminosity increases: the peak of the emission would change from the direction parallel (a pencil beam) to the direction perpendicular to B -field (a fan beam) with increasing luminosity. The fundamental lines become shallower as the viewing angle becomes larger, as shown in Figure 6. On the other hand, the second harmonic line tends to appear for higher luminosity because of a $\sin^{2(n-1)}\theta$ factor in the scattering cross-section. The deep second harmonic line would imply a fan-like beam pattern. In addition, the pulse profile tends to be complex but becomes double peaked in higher energies around the fundamental cyclotron energy for higher luminosity, while it tends to be a single peak for lower luminosity (Bildsten et al. 1997). This trend of the pulse profile seems to be consistent with the change from a pencil-like to a fan-like beam pattern with increasing luminosity. The ratio $E_{\text{cycl},2}/E_{\text{cycl},1}$ tends to be more than 2 (Caballero et al. 2007). In addition, $W_{1\text{st}} \gtrsim W_{2\text{nd}}$. Thus, the characteristics of the observed cyclotron lines agree with my numerical results as a result of the superposition of a large number of lines in a large polar cap region, as demonstrated in Section 4.1.

In the main pulse of the phase-resolved spectra (Caballero 2009), $W_{1\text{st}} \gtrsim W_{2\text{nd}}$ in certain phases of the main pulse with broader line width. In all phases of the main pulse, the harmonic ratios $E_{\text{cycl},2}/E_{\text{cycl},1}$ is more than 2. In addition, the observed energy of the broader line tends to be lower (Caballero 2009) and its depth tends to be deeper. Furthermore, the position of the lowest E_{cycl} can deviate somewhat from the peak of the main pulse. Thus, the fan-like beam is roughly able to explain the relationship between E_{cycl} and the intensity as well. These characteristics would imply that the superposition of a large number of lines occurs. A large polar cap region

$R_p \sim 2 \times 10^5$ cm demonstrated in Section 4.2 is able to reproduce the characteristics of the observed cyclotron lines in A0535+26.

Vela X-1. In the case of Vela X-1 with pulsation ~ 283 s, the luminosity reaches $\sim 10^{37}$ erg s $^{-1}$ (Lutovinov & Tsygankov 2009). The fundamental line tends to be very shallow and narrow with a deeper second harmonic line. This would imply that the viewing angle is larger. Moreover, the pulse profile indicates clear double peaks in higher energies around the fundamental cyclotron energy (e.g., Figure 4, Kreykenbohm et al. 2002). A fan-like beam is thus expected by these characteristics.

Kreykenbohm et al. (1999, 2002) reported that the observed harmonic ratio $E_{\text{cycl},2}/E_{\text{cycl},1}$ tends to be larger than 2 in the phase-resolved spectra of Vela X-1. The observed harmonic ratio is consistent with my numerical results described in Section 4.2. Next, in some observed spectra, the fundamental line can sometimes be too shallow to detect it as a line while the second harmonic line is relatively deep, so that only the second harmonic line can be detected (e.g., Orlandini et al. 1998). This is consistent with numerical results shown by the solid line in Figure 2 where the shallower fundamental and the deeper second harmonic lines are formed. In the ascent of the main pulse E_1 (Figure 7, Kreykenbohm et al. 2002) the deepest fundamental line was observed, where the line width seems to be relatively broad. These characteristics of the line would imply looking at a near magnetic pole (pole1) at relatively small viewing angle in which the fundamental line becomes deeper, as seen in Figure 6. Moreover, $W_{1\text{st}} \gtrsim W_{2\text{nd}}$ at most of pulse phases (Kreykenbohm et al. 2002), in contradiction to the relationship $W_{2\text{nd}} \sim 2W_{1\text{st}}$ predicted by theoretical Doppler broadening (Equation (11)). These relationships between the fundamental and the second harmonic in the line width and depth agree well with my numerical results described in Section 4.2.

In the main pulse, the observed energy of broader line tends to indicate lower energy (La Barbera et al. 2003), although the trend is not necessarily required in Kreykenbohm et al. (2002). Moreover, the fan-like beam is roughly able to explain the relationship between E_{cycl} and the intensity as well. In addition, the depth of the second harmonic line can be used to estimate the viewing angle. The deepest second harmonic line is in the descending edge of the main pulse, while the second harmonic line becomes less deep in the peak of the main pulse (Kreykenbohm et al. 2002). Furthermore, the shallowest second harmonic line is in the pulse minima. This trend is consistent with the results expected from the fan-like beam in pole1, in which the position of the deepest second harmonic line will deviate from the peak of the main pulse, whereas the position of the shallowest line will deviate as well. It can however become difficult to explain the relationship between E_{cycl} and the intensity in some phases even in the main pulse by my fan-like beam model, probably because the emission coming from a far magnetic pole (pole2) can begin to appear in the main pulse as luminosity increases. I will discuss these issues in Paper II.

Thus, the characteristics of the cyclotron lines in Vela X-1 as well as in A0535+26 would favor the line-forming region with a large polar cap region as demonstrated in Section 4.2. In this case, the line-forming region would be larger, because the shock height is expected to be on the same order of polar cap radius R_p around the critical luminosity given by Equation (12). Figure 8 illustrates the dimensions of the line-forming regions and the angular distribution of emissions for different sizes of the polar cap region. Here, the fundamental line tends to be formed around a higher site at the larger line-forming region,

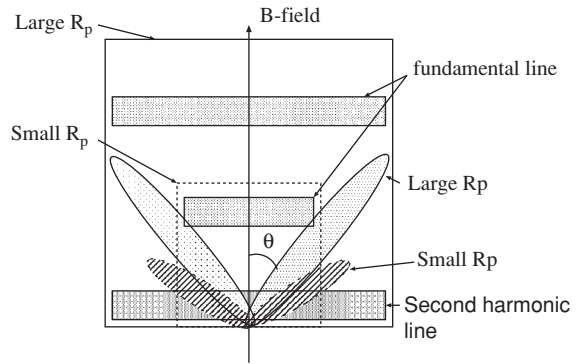


Figure 8. For a given luminosity, “Small R_p ” denotes a smaller polar cap region, while “Large R_p ” denotes a larger polar cap region. Dotted and solid square regions indicate line-forming regions for a smaller and a larger polar cap region, respectively. The line-forming region becomes larger and the direction of the peak emission θ becomes smaller as the size of polar cap region becomes larger. The fundamental line is formed around higher sites of the line-forming region, while the second harmonic line is formed around the bottom of the line-forming region.

while the second harmonic line is formed primarily around the bottom, as illustrated in Figure 4. As a result, the harmonic ratio $E_{\text{cycl},2}/E_{\text{cycl},1}$ is expected to become larger than 2 as the polar cap radius becomes larger. Thus, it is possible that the polar cap radius in Vela X-1 is smaller than that in A0535+26, considering that the observed values of the harmonic ratios of the line energies in Vela X-1 tends to be smaller than those in A0535+26. If so, the critical luminosity given by Equation (12) in Vela X-1 would be smaller than that in A0535+26. As a result, a fan-like beam can be formed in spite of smaller luminosity in Vela X-1 than in A0535+26, as illustrated in Figure 8. In fact, the pulse profile around the fundamental cyclotron energy in Vela X-1 appears to be narrower than that in A0535+26 possibly due to a smaller emission region.

GX 301-2. In the case of GX 301-2 with pulsation ~ 681 s, the fundamental line is even shallower. A maximum luminosity is $\sim 2 \times 10^{37}$ erg s $^{-1}$ (La Barbera et al. 2005). The pulse profile indicates relatively double peaks in higher energies. However, the second harmonic line has not been detected yet. Moreover, in the main pulse, the fan-like beam does not seem to be able to explain the relationship between E_{cycl} and the intensity. In addition, E_{cycl} is observed to increase with increasing luminosity. I will discuss this source in detail in a forthcoming paper.

5.1.2. Relatively Lower Luminosity $\sim 10^{36}$ erg s $^{-1}$

I will consider accreting X-ray pulsars with a relatively lower luminosity $\sim 10^{36}$ erg s $^{-1}$, which is expected to be less than L_{crit} . In this case, the angle of the peak emission is expected to become smaller, since a beam pattern would approach a pencil-like beam, as shown in Figure 7. As a result, the fundamental line would become deeper and broader in Figure 6, while the second harmonic line would become somewhat shallower. In the pulse profiles of these pulsars, it is likely that the secondary peak tends to weaken due to the angular distribution of the emission with a peak at a relatively small angle.

4U 1538-52. In the case of 4U 1538-52 with pulsation ~ 529 s, the fundamental line is relatively shallow, but deeper and broader than in Vela X-1 (Clark et al. 1990). A relatively deep second harmonic line has been detected at ~ 47 keV using data from *RXTE* and *INTEGRAL* (Rodes-Roca et al. 2009). The harmonic ratio $E_{\text{cycl},2}/E_{\text{cycl},1}$ is reported to be ~ 2.2 . The

secondary pulse is even weaker than the main pulse. In the main pulse of 4U 1538–52, a pencil-like beam is roughly able to explain the relationship between E_{cycl} and the intensity in the main pulse, considering an angular distribution of the intensity as illustrated by the dotted line in Figure 7. This pencil-like beam pattern is similar to that implied from the features observed in 4U 1538–52 (Clark et al. 1990). Nagel (1981a) also shows an analogous angular dependence of the intensity of the escaping radiation from a radiating slab geometry, solving two coupled diffusion equations for extraordinary and ordinary photons. This angular distribution in the line-forming region with dipole B -field leads to higher line energy for a narrower line. Furthermore, the position of the lowest energy of the cyclotron lines in the pulse profile would deviate somewhat from the peak intensity. This angular distribution model is able to explain the characteristics of the observed line energy, except for those of the broadest line. In the case of the observed broadest line, it does not have the lowest energy but a somewhat higher energy, which agrees with the numerical results in Section 4.2. In addition, even in the secondary pulse, a pencil-like beam may also be able to be applied.

4U 1907+09. In the case of 4U 1907+09 with a pulsation ~ 440 s, a shallower fundamental line ~ 18.8 keV and deeper second harmonic line ~ 39.4 keV were detected in phase-averaged spectrum with *BeppoSAX* (Cusumano et al. 1998). The harmonic ratio $E_{\text{cycl},2}/E_{\text{cycl},1}$ is reported to be ~ 2.1 , which could imply a smaller line-forming region than in 4U 1538–52. The polar cap radius could therefore be smaller as well, as shown in Figure 8. In this case, a pencil-like beam is expected to become somewhat closer to a fan-like beam. The secondary pulse is also even weaker than the main pulse in higher energies (17–36 keV), while it appears to be comparable to the main pulse in lower energies (2–17 keV). Makishima et al. (1999) also reported that the phase-resolved spectra show deeper fundamental lines at some phases with *Ginga*. In addition, the fundamental line is deeper and broader than that in Vela X–1. These characteristics are consistent with those of the cyclotron line expected in a pencil-like beam.

MXB 0656–072. In the case of MXB 0656–072 with a pulsation ~ 160 s, relatively shallow and rather broad lines were observed, showing little variability in E_{cycl} with pulse phase in comparison with some other sources (McBride et al. 2006). The second harmonic line is not clearly visible. The observed pulse profile is one broad pulse, which seems to be characterized by a pencil beam pattern. This implies that one may view only a single magnetic pole as pointed out in McBride et al. (2006). The observed line width simply increases with pulse phase during the one broad pulse. This would correspond to a variation in the viewing angles from a larger to a smaller angle in a pencil-like or a fan-like beam, because in a pencil or a fan beam pattern the line width is expected to increase and then decrease or decrease and then increase, respectively, during one pulse, i.e., the increase and decrease of the intensity. In this source, the polar cap radius is possibly larger, because a pencil-like beam is expected despite the somewhat larger luminosity, $L = (6.6 \pm 0.4) \times 10^{36}$ erg s $^{-1}$. In addition, such a larger polar cap region naturally leads to a broader and shallower fundamental line. Thus, it is likely that the beam pattern in MXB 0656–072 is a pencil-like beam.

5.1.3. Lower Luminosity $\sim 10^{34}$ – 10^{35} erg s $^{-1}$

I consider accreting X-ray pulsars with so low luminosity $\sim 10^{34}$ – 10^{35} erg s $^{-1}$ which would be much less than L_{crit} . In this case, the angle of the peak emission is expected to become

much smaller. As a result, the fundamental line would become deeper and broader in Figure 6, while the second harmonic line would become much shallower or disappear. In the pulse profiles of these pulsars, secondary peak would tend to be quite small due to the angular distribution of the emission with a peak at a small angle. For such a low luminosity $\sim 10^{34}$ – 10^{35} erg s $^{-1}$, the number of the superposed lines may however decrease because of the lower shock height. Even in this case, the fundamental line remains relatively deep and broad at the smaller viewing angle that is applied to sources discussed in this section. These pulsars would therefore be able to be discussed with my model.

4U 0352+309. In the case of 4U 0352+309 with pulsation ~ 837 s, the deepest fundamental lines were observed in accreting X-ray pulsars with long spin period (Coburn et al. 2001). This is likely because the luminosity is so low, $\sim 10^{34}$ erg s $^{-1}$, that a pencil beam is expected. Therefore, the angle of the peak emission with respect to B -field will be so small that the cyclotron line tends to become broader and deeper as can be seen at an angle of $\mu = 0.96$ in Figure 6. The pulse profile is also one broad pulse, which is consistent with a pencil beam pattern.

A0535+26. In the case of A0535+26 with lower luminosity $\sim 10^{35}$ erg s $^{-1}$, the observed energy of the cyclotron line seems to be constant but its depth seems to change. The fundamental line tends to be deeper, while the second harmonic feature at ~ 100 keV was not required (Terada et al. 2006). These characteristics agree with those at an angle of $\mu = 0.96$ in Figure 6. Furthermore, the pulse profile is more or less one broad pulse, which is consistent with a pencil beam pattern.

5.2. Relatively Long Spin Periods (~ 10 – 100 s)

I consider the property of the cyclotron lines in several accreting X-ray pulsars with relatively long spin periods (~ 10 – 100 s) in terms of my model, although Arons & Lea's (1980) model may not be able to be applied.

OAO 1657–415. In the case of OAO 1657–415 with a pulsation ~ 38 s, Orlandini et al. (1999) discussed the presence of very broad and shallow fundamental lines, while they concluded that no cyclotron line is needed. Barnstedt et al. (2008) also did not find an indication for a cyclotron line both in phase-resolved and phase-averaged spectra. It is possible that the fundamental line is too shallow for it to be detected as a line. The luminosity is expected to be as high as in Vela X–1, or the polar cap radius may be somewhat smaller.

CepX–4. In the case of CepX–4 with a pulsation ~ 66 s, relatively deep and broad lines were observed despite a relatively long spin period (Mihara et al. 1991; McBride et al. 2007). In this case, the angle of the peak emission with respect to B -field is expected to be relatively small so that the fundamental line is broader and deeper in terms of my model. The luminosity is therefore expected to be relatively low or the polar cap radius may be relatively large. In the main pulse, the pencil-like beam is roughly able to explain the relationship between E_{cycl} and the intensity, although the behaviors of the energy-dependent pulse profiles appear to be analogous to those of 4U 1907+09 described above.

EXO 2030+375. In the case of EXO 2030+375 with a pulsation ~ 42 s, a possible cyclotron feature at ~ 36 keV has been reported during a normal outburst with *RXTE* (Reig and Coe 1999) at lower luminosities ($L < 5 \times 10^{36}$ erg s $^{-1}$). On the other hand, the evidence of an absorption line at ~ 63 keV has been shown during a giant outburst (Klochkov et al. 2008). Klochkov et al. (2008) interpreted that the features

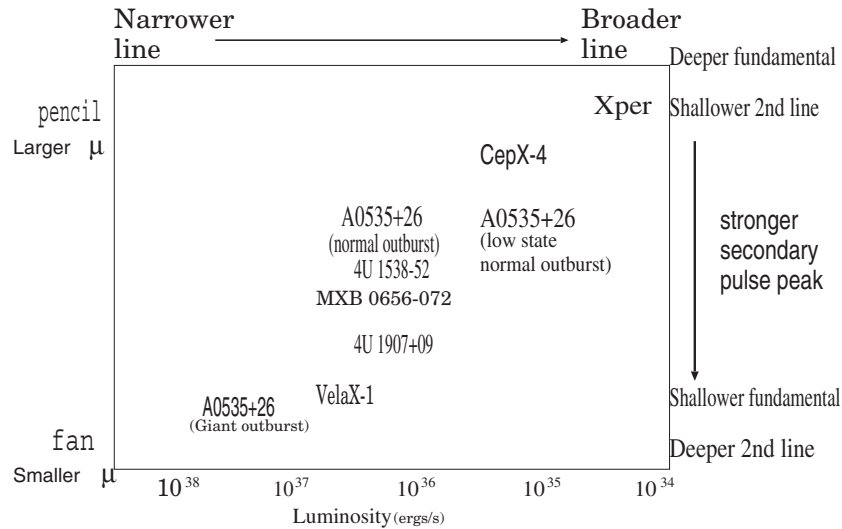


Figure 9. Distribution of accreting X-ray pulsars for the expected angle of the peak emission as a function of luminosity in terms of my model. As the luminosity increases, the fundamental lines tend to be narrower and shallower. In contrast, higher harmonic lines tend to be deeper. In addition, the peak of the secondary pulse profile tends to be stronger.

at ~ 36 keV and at ~ 63 keV are the fundamental and the second harmonic lines, respectively. My model implies that the features at ~ 36 keV and at ~ 63 keV would, respectively, be deeper and even shallower for lower luminosity, which corresponds to a normal outburst. Meanwhile, the features at ~ 36 keV and at ~ 63 keV would, respectively, be even shallower and deeper for higher luminosity, which correspond to a giant outburst. As a result, only a feature at ~ 36 keV has been detected during a normal outburst, while only a feature at ~ 63 keV during a giant outburst. Thus, my model is able to explain the variation on the depth of both the fundamental line and the second harmonic between a normal and a giant outburst.

Thus, even in the accreting X-ray pulsars with relatively long spin periods (~ 10 – 100 s), some characteristics of the observed cyclotron lines seem to be able to be explained by my present model.

5.3. Comprehensive Comparisons

I plot accreting X-ray pulsars for the expected angle of the peak emission with respect to the B -field as a function of luminosity in terms of my model in Figure 9. In Table 3, spin periods and the parameters of observed cyclotron lines, including those for pulsars with short spin periods for comparison, are shown. The parameters of observed cyclotron lines at the fundamental and the second harmonic in accreting X-ray pulsars with long spin periods in which the second harmonic line was detected are also summarized in Table 4. This plot is similar to the relationship between the depth and the width in the observed cyclotron lines (Coburn et al. 2002). Thus, the depth and the width in the observed cyclotron lines can be related to the luminosity as well as the viewing angle. As the luminosity increases, the X-ray beam pattern is expected to change gradually from a pencil-like to a fan-like beam pattern, i.e., the angle of the peak emission would become larger. For accreting X-ray pulsars with higher luminosity, the fundamental lines thus become narrower due to a larger viewing angle and shallower as a result of the superposition of the lines, while higher harmonic lines tend to be also narrower but remain deeper as described in Section 4.1. In addition, in observations, the peak of the secondary pulse tends to be stronger with increasing luminosity. The change in the X-ray

Table 3
Observed Spin Periods, Cyclotron Line Energy, Width, and Depth

Object	P_{spin} (s)	E_c (keV)	W (keV)	D
X Per ^a	837.7	$28.6^{+1.5}_{-1.7}$	9.0 ± 1.3	0.66 ± 0.12
GX301–2 ^a	681	$42.4^{+3.8}_{-2.5}$	$8.0^{+1.8}_{-2.6}$	$0.5^{+0.3}_{-0.2}$
4U1538–52 ^a	528.8	$20.66^{+0.05}_{-0.06}$	2.15 ± 0.04	0.50 ± 0.01
4U1907+09 ^a	440.4	18.3 ± 0.4	1.6 ± 0.2	0.26 ± 0.04
VelaX–1 ^a	283.2	$24.4^{+0.5}_{-1.1}$	$0.9^{+0.9}_{-0.8}$	$0.16^{+1.13}_{-0.07}$
MXB 0656–072 ^b	160.4	$32.8^{+0.5}_{-0.4}$	$11.8^{+1.0}_{-1.1}$	$0.38^{+0.02}_{-0.02}$
A0535+26 ^c	103.39315	45.9 ± 0.3	9.5 ± 0.3	0.415 ± 0.015
A0535+26 ^d	103.375	$45.5^{+1.3}_{-1.3}$	$10.9^{+4.2}_{-2.9}$	$1.8^{+0.4}_{-0.3}$
Cep X–4 ^e	66.2490	$30.7^{+1.8}_{-1.9}$	$3.6^{+2.9}_{-1.5}$	$0.7^{+0.3}_{-0.2}$
4U1626–67 ^a	7.67	$39.3^{+0.6}_{-1.1}$	$6.6^{+0.6}_{-0.7}$	2.1 ± 0.2
CenX–3 ^a	4.82	$30.4^{+0.3}_{-0.4}$	7.1 ± 0.2	1.08 ± 0.02
X0115+63 ^a	3.61	$16.4^{+0.4}_{-1.0}$	$3.6^{+0.5}_{-1.0}$	$0.78^{+0.07}_{-0.22}$
HerX–1 ^a	1.2377	$40.4^{+0.8}_{-0.3}$	$6.4^{+1.1}_{-0.4}$	$0.76^{+0.10}_{-0.08}$

Notes.

^a Coburn et al. (2002).

^b McBride et al. (2006).

^c Caballero et al. (2007; *INTEGRAL*).

^d Terada et al. (2006).

^e McBride et al. (2007).

beam pattern from a pencil-like to a fan-like beam pattern would contribute to the variation on the peak intensity of the secondary pulse.

6. CONCLUSION AND SUMMARY

The present model is revised by including the superposition of a large number of lines and the gradient in the density and temperature of an electron–proton plasma in Nishimura (2005). This present model would be appropriate for X-ray accreting pulsars with luminosity $\lesssim 10^{36}$ – 10^{37} erg s $^{-1}$ because of the cylindrical plane-parallel geometry as described in Figure 1. On the other hand, Nishimura’s (2008) model would be able to be applied to X-ray accreting pulsars with luminosity $\gtrsim 10^{36}$ – 10^{37} erg s $^{-1}$ because of the cylindrical geometry with an emission region along the cylindrical axis. In this paper, I discussed the

Table 4
Observed Cyclotron Line Energy, Width, and Depth of the Fundamental and the Second Harmonic

Object	$E_{\text{cyc},1}$ (keV)	$E_{\text{cyc},2}$ (keV)	W_1 (keV)	W_2 (keV)	D_1	D_2
VelaX–1 ^a	$23.3^{+1.3}_{-0.6}$	$54.5^{+5.0}_{-2.8}$	$4.9^{+1.9}_{-0.8}$	$5.3^{+5.1}_{-2.8}$	$0.15^{+0.07}_{-0.02}$	$0.5^{+0.3}_{-0.1}$
A0535+26 ^b	45.9 ± 0.3	102^{+4}_{-3}	9.5 ± 0.3	8^{+3}_{-2}	0.415 ± 0.015	$1.1^{+0.4}_{-0.3}$
4U1538–52 ^c	$21.4^{+0.9}_{-2.4}$	$47.1^{+2.2}_{-1.7}$	3^{+4}_{-2}	$5^{+5}_{-2.3}$	$0.43^{+0.3}_{-0.07}$	$0.88^{+0.5}_{-0.24}$
4U1907+09 ^d	18.8 ± 0.4	39.4 ± 0.6	2.2 ± 0.4	3.6 ± 0.7		

Notes.

^a Kreykenbohm et al. (2002; Main Pulse Center).

^b Caballero et al. (2007; *INTEGRAL*).

^c Rodas-Roca et al. (2009).

^d Cusumano et al. (1998).

properties of the cyclotron lines in accreting X-ray pulsars with long spin periods (~ 100 – 800 s). My simplified model succeeds in reproducing the asymmetric line profile, the anharmonic line energy ratios $E_{\text{cyc},2}/E_{\text{cyc},1} > 2$, and the shallower fundamental line feature with deeper second harmonic, as often seen in accreting X-ray pulsars with long spin periods. Moreover, the correlations between $E_{\text{cyc},1}$ and the intensity of the pulse profile can be explained in terms of my model. I also show that the fundamental line tends to be even shallower and narrower for higher luminosity, probably because the direction of the peak of the emission approaches the direction normal to the B -field. In addition, the fundamental line becomes shallower via the superposition of a large number of lines in a larger polar cap region such as $R_p \sim 2 \times 10^5$ cm. My model therefore implies that the polar cap region is larger in accreting X-ray pulsars with a long spin period, while it is relatively small in those with a short spin period. In addition, in a large polar cap region, a line-forming region with lower density would be formed, since the density is proportional to $L_x/\pi R_p^2$. Consequently, the fundamental line tends to be shallower.

It is possible that this larger line-forming region, where the superposition of a large number of lines operates, is formed by a large-scaled radiation-supported scattering atmospheres (Sturmer & Dermer 1994), a large-scaled accretion halo (thin gas region outside an accretion column), or a free-fall region as well as a sinking region below a shock in the accretion column with a larger polar cap region. These regions can be a candidate for a relatively thin and extended line-forming region in my model under a dipole magnetic field configuration. Moreover, the 1–0 geometry I use in this work is suitable for a line-forming region such as a radiation-supported scattering atmosphere in a magnetosphere (e.g., Isenberg et al. 1998). It should however be noted that the 1–0 geometry would be able to be applied to an accretion column if the photons that emerge from the bottom of the slab return to the stellar surface where they are thermalized. The radiation-supported scattering atmosphere is sometimes correlated with luminosity, because the size of the atmosphere depends on the radiation spectrum emitted from a polar cap region and the availability of accreting material to sample the regions of the NS magnetosphere outside an accretion column. The radiation-supported scattering atmosphere is considered to be formed in a region surrounding an accretion column if there is a weak flow of matter into this region or cross-field line spatial diffusion of matter.

An alternative interpretation of a larger number of superposing lines with different line energies can be a contribution from other polar cap region, such as a gravitational light bending or the deviation from antipodality of the caps (Ruderman 1991) or a contribution from the presence of high-order multipoles, a

random or chaotic magnetic field. The presence of a non-dipole B -field (Gil et al. 2002; Nishimura 2005) or high-order multipoles are able to yield a large variation on the strength of the magnetic field in a small region $\lesssim 10^4$ cm. These phenomena will also be able to create a large variation in the line energy at the second harmonic as well as the fundamental.

In the present paper, I discussed the characteristics of the lines in the line-forming region with decreasing B -field, considering the dipole B -field. In Paper II, I will consider cyclotron lines forming in the line-forming region with increasing B -field, improving Nishimura (2005) by including the superposition of a large number of lines and the gradient in the density and temperature of an electron–proton plasma.

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REFERENCES

- Araya, R. A., & Harding, A. K. 1999, *ApJ*, **517**, 334
 Araya-Góchez, R. A., & Harding, A. K. 2000, *ApJ*, **544**, 1067
 Arons, J., & Lea, S. M. 1980, *ApJ*, **235**, 1016
 Barnstedt, J., et al. 2008, *A&A*, **486**, 293
 Becker, P. A. 1998, *ApJ*, **498**, 79
 Becker, P. A., & Wolff, M. T. 2005, *ApJ*, **630**, 465
 Bildsten, L., et al. 1997, *ApJS*, **113**, 367
 Bulik, T., Mészáros, P., Woo, J. W., Hagase, F., & Makishima, K. 1992, *ApJ*, **395**, 564
 Bulik, T., Riffert, H., Mészáros, P., Makishima, K., Mihara, T., & Thomas, B. 1995, *ApJ*, **444**, 405
 Caballero, I. 2009, PhD thesis, Univ. Tübingen
 Caballero, I., et al. 2007, *A&A*, **465**, L21
 Clark, G. W., Woo, J. W., Nagase, F., Makishima, K., & Sakao, T. 1990, *ApJ*, **353**, 274
 Coburn, W., Heindl, W. A., Gruber, D. E., Rothschild, R. E., Staubert, R., Wilms, J., & Kreykenbohm, I. 2001, *ApJ*, **552**, 738
 Coburn, W., Heindl, W. A., Rothschild, R. E., Gruber, D. E., Kreykenbohm, I., Wilms, J., Kretschmar, P., & Staubert, R. 2002, *ApJ*, **580**, 394
 Cusumano, G., di Salvo, T., Burderi, L., Orlandini, M., Piraino, S., Robba, N., & Santangelo, A. 1998, *A&A*, **338**, L79
 Gil, J. A., Melikidze, G. I., & Mitra, D. 2002, *A&A*, **388**, 235
 Grove, J. E., et al. 1995, *ApJ*, **438**, L25
 Harding, A. K., & Daugherty, J. K. 1991, *ApJ*, **374**, 687
 Isenberg, M., Lamb, D. Q., & Wang, J. C. L. 1998, *ApJ*, **493**, 154
 Klochkov, D., Santangelo, A., Staubert, R., & Ferrigno, C. 2008, *A&A*, **491**, 833
 Kreykenbohm, I., Coburn, W., Wilms, J., Kretschmar, P., Staubert, R., Heindl, W. A., & Rothschild, R. E. 2002, *A&A*, **395**, 129
 Kreykenbohm, I., Kretschmar, P., Wilms, J., Staubert, R., Kendziorra, E., Gruber, D. E., Heindl, W. A., & Rothschild, R. E. 1999, *A&A*, **341**, 141

- La Barbera, A., Santangelo, A., Orlandini, M., & Segreto, A. 2003, *A&A*, **400**, 993
- La Barbera, A., Segreto, A., Santangelo, A., Kreykenbohm, I., & Orlandini, M. 2005, *A&A*, **438**, 617
- Lamb, D. Q., Wang, J. C. L., & Wasserman, I. 1990, *ApJ*, **363**, 670
- Lutovinov, A. A., & Tsygankov, S. S. 2009, *Astron. Lett.*, **35**, 433
- Lyubarskii, Y. E., & Syunyaev, R. A. 1988, *Sov. Astron. Lett.*, **14**, 390
- Maisack, M., Grove, J. E., Kendziorra, E., Kretschmar, P., Staubert, R., & Strickman, M. S. 1997, *A&A*, **325**, 212
- Makishima, K., Mihara, T., Nagase, F., & Tanaka, Y. 1999, *ApJ*, **525**, 978
- McBride, V. A., et al. 2006, *A&A*, **451**, 267
- McBride, V. A., et al. 2007, *A&A*, **470**, 1065
- Mészáros, P., & Nagel, W. 1985, *ApJ*, **298**, 147
- Mihara, T., Makishima, K., Kamiyo, S., Ohashi, T., Nagase, F., Tanaka, Y., & Koyama, K. 1991, *ApJ*, **379**, 61
- Nagel, W. 1981a, *ApJ*, **251**, 278
- Nagel, W. 1981b, *ApJ*, **251**, 288
- Nakajima, M., Mihara, T., Makishima, K., & Niko, H. 2006, *ApJ*, **646**, 1125
- Nishimura, O. 2003, *PASJ*, **55**, 849
- Nishimura, O. 2005, *PASJ*, **57**, 769
- Nishimura, O. 2008, *ApJ*, **672**, 1127
- Orlandini, M., dal Fiume, D., del Sordo, S., Frontera, F., Parmar, A. N., Santangelo, A., & Segreto, A. 1999, *A&A*, **349**, L9
- Orlandini, M., et al. 1998, *A&A*, **332**, 121
- Poutanen, J., & Gierliński, M. 2003, *MNRAS*, **343**, 1301
- Reig, P., & Coe, M. J. 1999, *MNRAS*, **302**, 700
- Rodes-Roca, J. J., Torrejón, J. M., Kreykenbohm, I., Martínez Núñez, S., Camero-Arranz, A., & Bernabéu, G. 2009, *A&A*, **508**, 395
- Ruderman, M. 1991, *ApJ*, **366**, 261
- Schönherr, G., Wilms, J., Kretschmar, P., Kreykenbohm, I., Santangelo, A., Rothschild, R. E., Coburn, W., & Staubert, R. 2007, *A&A*, **472**, 353
- Sturmer, S. J., & Dermer, C. D. 1994, *A&A*, **284**, 161
- Terada, Y., et al. 2006, *ApJ*, **648**, L139
- Wang, J. C. L., Wasserman, I., & Lamb, D. Q. 1993, *ApJ*, **414**, 815
- Wang, J. C. L., Wasserman, I., & Salpeter, E. E. 1988, *ApJS*, **68**, 735
- Wang, J. C. L., Wasserman, I., & Salpeter, E. E. 1989, *ApJ*, **338**, 343